IP DAY 2026

DETAILED GUIDELINES

Line following robot

FIELD SPECIFICATION

Path: A black winding line (~2 to 3 cm width) representing the route.
Field Size: 3m x 5m rectangle
 Zones: o Start Zone – Robot placement point.
Fork Zone (x2): Multiple path branches, only one correct.

Note: The field is bordered; robots going out of bounds are disqualified.

POINTS

- Parameter Weightage
- Task Completion 30%
- Time Efficiency 20%
- Navigation Accuracy 20%
- Obstacle Handling 20%
- Innovation in Design 10%

CONTACT

+91 8849930780
kirtan.robodev@gmail.com
Government Polytechnic Ahmedabad, Dr Vikram Sarabhai Marg, Panjara Pol, Ambawadi, Ahmedabad, Gujarat 380015

https://maps.app.goo.gl/VEQJTP4tNcQue2bY8t

TASKS TO DO

- 1.Speed and Smoothness
- 2.The S-Curve
- 3.The Speed Bump
- 4.The Line Gap
- 5.90 degree turn
- 6.cross junction

ROBOT SPECIFICATIONS

- Must be pre-built, autonomous
- Self-powered (battery only); no combustion engines allowed
- No external controllers or live intervention
- No restriction on power/voltage as long as safety standards are meet
- Maximum Dimensions: 18 cm (W) × 20 cm (L) × 25 cm (H)
- Sensors Allowed: o Camera module o IR/Ultrasonic sensors for obstacle detection o Line-following sensors
- Autonomy: Fully autonomous (no remote control or external guidance)

SAFETY AND CONDUCT

- Safety and Conduct: Robots must not cause damage to the field or audience.
- Judges' decision will be final in case of disputes or technical queries.